Physics of the Earth and Planetary Interiors, 7 (1973) 213–216
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RESEARCH NOTES

NOTE ON THE PALAEOMAGNETISM OF THE SEKONDI SERIES, GHANA

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Accepted for publication 16 November 1972

From a collection of 39 oriented hand-samples at 16 sites, total N.R.M. directions at 12 sites from the Elmina Sandstone (Devonian or possibly Carboniferous) of the Ghana coast fall in a group. Their in-situ mean ($D = 334°$, $I = +11\frac{1}{2}°$) is significantly divergent from the local geomagnetic field, and does not correlate with expected Palaeozoic remanence directions. A bedding-tilt test suggests that the magnetisation is secondary, and comparison with other African data suggests a Mesozoic (possibly Cretaceous) age. The remanence is only partially stable against thermal demagnetisation. The observations are consistent with a remanence originating at the time of faulting, tilting and uplift which marked the beginning of rifting of South America from Africa.

1. Introduction

This small study is one of several recent attempts to remedy the almost non-existent palaeomagnetic record for West Africa. The rocks involved are sandstones of Late Palaeozoic age on the coast of Ghana, at approximately 5°N, 1.3–2.3°W. A total of 39 hand-samples from 16 sites were collected at various times by two of us (W.R.F. and M.W.M.) and R.M. Shackleton. They were oriented by magnetic compass and clinometer which is an adequate method in this area. Total N.R.M. measurements were made in Leeds by J.C.B. and J.T. Sallomy. Thermal demagnetisation of 29 samples from 15 sites was carried out in Canberra (M.W.M.).

The Sekondi Series has been comprehensively mapped and described by Crow (1952) and consists in order of age of the Ajua Shales (45–65 m), Elmina Sandstone (300–400 m), Takoradi Beds (400 m), Efia Nkwanta Beds (130 m) and, at the top of the succession, the Sekondi Sandstone (about 300 m). The Elmina Sandstone was the principal object of the present study, and sampling was concentrated in two areas, along 3 km of the coast eastward from Elmina where the beds dip moderately to the south, and within 5 km north and west of Takoradi where they are more or less horizontal. Most of the samples are of chocolate and purple feldspathic and micaceous sandstone, the coarser beds being avoided. The only other formation visited was the Efia Nkwanta Beds where two samples were taken near the type locality. These showed widely divergent directions of magnetisation so the remainder of this discussion is confined to the Elmina Sandstone.

Age control on the Sekondi Series is limited. Fossils are confined to the Takoradi Beds immediately overlying the Elmina Sandstone. Crow (1952) reports the occurrence of Mollusca and Brachiopoda which Cox (1946) assigns to the Devonian, though not excluding the possibility of a rather younger, Carboniferous age. There is no angular discordance between the Elmina Sandstone and the Takoradi Beds and sedimentation appears to be continuous in most places

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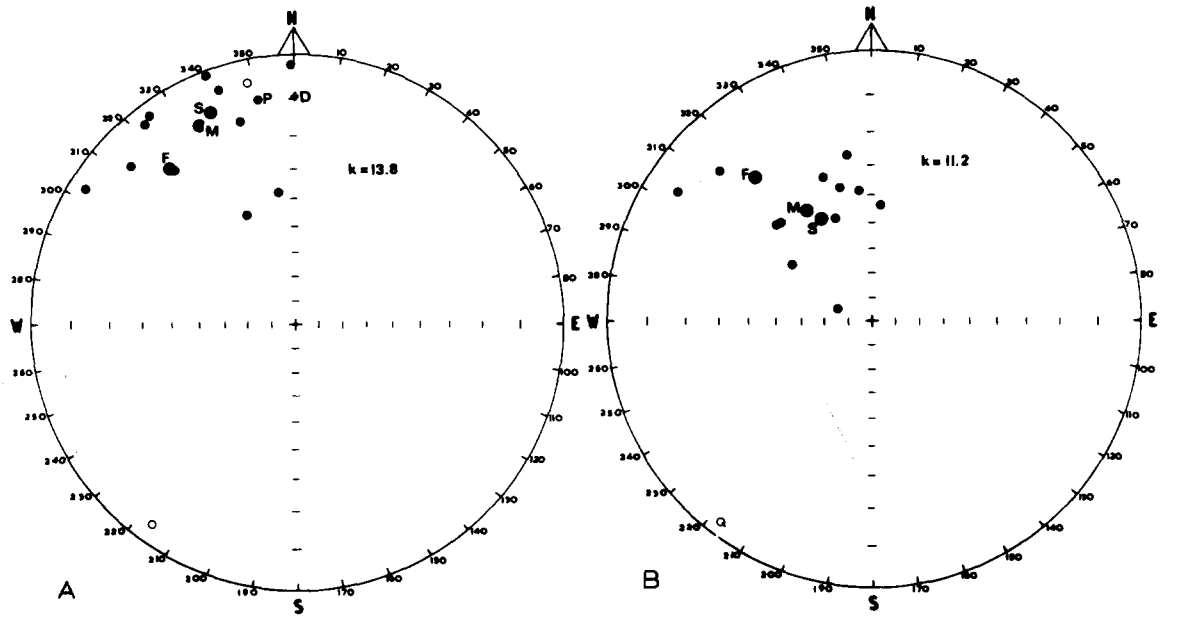


Fig. 1. Stereographic projection of site mean directions of total N.R.M. (smaller circles). The larger circles show the overall mean (*M*) of the twelve grouped sites, the mean of the flat-lying sites (*F*) and the mean of the southward dipping sites (*S*). A: in situ; B: dip corrected. *P* denotes the present geomagnetic field and *D* the geocentric axial dipole field at the sampling region. Open symbols denote upwards, solid symbols downwards.

where the passage beds outcrop, though in some localities breccias or conglomerates define the boundary. There is thus no reason to assign a significantly different age to the palaeomagnetically sampled horizons from that of the Takoradi Beds, so they may be regarded tentatively as Devonian. A Devonian age is also preferred on the somewhat tenuous grounds that Carboniferous rocks are unknown in this part of West Africa while established Devonian rocks occur near Accra.

2. Palaeomagnetic results

Total N.R.M. directions are summarised in Fig. 1. Two sites must be considered to be randomly magnetised on Watson's (1956) criterion. Apart from site 7, which is magnetised in a southwesterly direction, the magnetisations at the other twelve sites form a group (Fig. 1). Site 7 has been excluded from further discussion. The in-situ mean direction, the dip-corrected mean, the local geomagnetic field and the geocentric axial dipole field at 5°N are all significantly different from each other at the 95% confidence level. Thus the remanence is not likely to be geologically recent. However, the dip-corrected mean conforms neither to the Devonian palaeomagnetic field for Africa inferred from studies in southern Africa, nor to any later inferred palaeomagnetic field. Moreover there are two indications that the remanence is more likely to be secondary. The means of the southward-dipping and flat-lying sections are 18° apart in situ but 27° apart when the beds are restored to the horizontal by simple rotation about the local strike. Also the precision of grouping at the southward-dipping sites is slightly better in situ than after dip correction (Table I). Although these differences are not statistically significant at the 95% probability level they do suggest that the remanence is younger than the tectonic deformation. An estimate of the age of this secondary N.R.M. can be obtained by comparing its palaeomagnetic pole with the present geomagnetic field and with the standard polar wander curve for southern Africa. Assuming no large-scale relative movement between Ghana and southern Africa since the deposition of the Elmina Sandstone, a Palaeozoic age of remanence is not likely

TABLE I
Elmina Sandstone overall statistics

		N	R	k	α_{95} (°)	Decl. (°)	Incl. (°)	Palaeomagnetic pole Lat. (°)	Long. (°)
Nearly flat-lying sites (numbers 6, 8, 9)	in situ	3	2.73	7.5	–	321	$+17\frac{1}{2}$	51N	276E
	dip corrected	3	2.75	8.0	–	321	+21	–	–
Southward-dipping sites (numbers 2, 4, 5, 10 – 15)	in situ	9	8.60	18.8	12.2	338	$+ 9\frac{1}{2}$	68N	268E
	dip corrected	9	8.47	15.1	13.7	334	+44	–	–
All grouped sites	in situ	12	11.20	13.8	12.1	334	$+11\frac{1}{2}$	64N	271E
	dip corrected	12	11.02	11.2	13.6	330	$+38\frac{1}{2}$	–	–

R = length of the resultant of the N site mean vectors;
k = precision estimate (Fisher, 1953);
α_{95} = semi-angle of cone of 95% confidence.

(Fig. 2). The concordance of this new result with Mesozoic poles in Fig. 2 may indicate a Mesozoic age for the magnetisation of the Elmina Sandstone. The problem of relative movement since the Mesozoic is unlikely to be serious both on plate-tectonic evidence and on palaeomagnetic grounds – Mesozoic poles from Morocco (Hailwood and Mitchell, 1971) lie in the same area as the southern African poles. In the absence of positive palaeomagnetic evidence, the precise age and mode of origin for the remanence of the Elmina Sandstone has to be deduced from the post-depositional history of the rocks.

Crow (1952) regarded the tilting of the beds as a consequence of the complex faulting which is characteristic of this part of the West African coast. Indirect evidence dates this faulting as probably pre-Late Cretaceous, and Crow (1952) follows Wegener (1924) and Du Toit (1937) in associating it with the rifting of South America from Africa. The opening of the South Atlantic is regarded as having started about 130 m.y. ago close to the Jurassic–Cretaceous boundary as judged by the earliest marine deposits on the western margin of Africa (Funnell and Smith, 1968) and off the eastern margin of South America (Fox et al., 1970). Results of deep-sea drilling (Maxwell et al., 1970) confirm this view. It is tempting therefore to assign both the faulting and magnetisation to an Early to Mid-Cretaceous age and to note

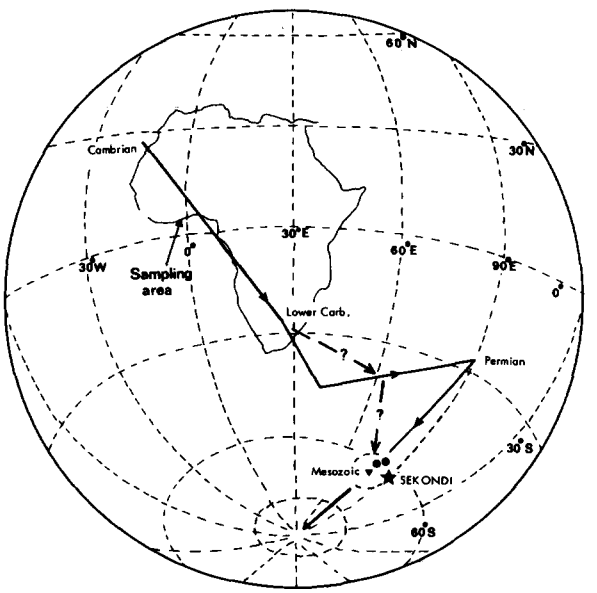


Fig. 2. Palaeomagnetic pole from the Elmina Sandstone, in situ (Table I) compared with the palaeomagnetic south polar wander curve for southern Africa (McElhinny et al., 1968). The mean Mesozoic pole is denoted by a triangle, and it is encircled by its 95% confidence circle. The solid dots denote Early to Mid-Cretaceous poles from Malawi (Briden, 1967) and Mozambique (Gough and Opdyke, 1963) with which the present result (denoted by a star) agrees closely.

the very close palaeomagnetic agreement with southern African poles of that age (solid circles in Fig. 2). The normal polarity of the Sekondi rocks is consistent with that found through most of the Cretaceous (Helsley and Steiner, 1969). Such a definite conclusion may not be fully justified however, since Early Mesozoic poles are scarcely distinguishable from the Cretaceous ones, and it is not known at what stage in the continental rifting the faulting occurred.

On this model the remanent magnetisation would have accumulated during the Mesozoic and stabilised during the uplift which commonly accompanies intracontinental rifting. It could be a C.R.M. stabilised by uplift and a change in groundwater regime which could end the crystallisation of the matrix of these sandstones. Alternatively it could be a viscous P.T.R.M. stabilised by the fall in temperature accompanying uplift (Chamalaun, 1964; Briden, 1965). The thermal-demagnetisation curves show a steady reduction in moment to about half of the total at 600°C. But no distinct high-temperature component emerges and the dispersion of directions becomes progressively rather worse with increasing temperature. Though this evidence is by no means conclusive, these characteristics are more commonly associated with viscous P.T.R.M. involving maximum temperatures of (say) 100°C than with C.R.M. which is commonly more stable against thermal cleaning.

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